

Day 11 Review Sheet · Days 5-10

MathCraft · Grade 6 Mathematics · Professor Xavier Honablue, M.Ed.

Name: _____ Date: _____ Hour: _____

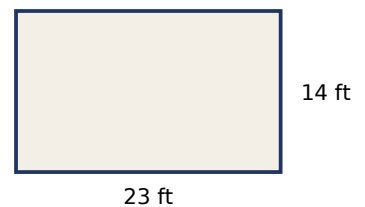
Covers: rectangle area & two-digit multiplication (Day 5) · missing sides (Day 6) · compound rectangles (Day 7) · parallelograms (Days 8-9) · multiplying decimals & fractions (Day 10). **Standards:** 6.G.A.1, 6.NS.B.2, 6.NS.B.3, with prerequisite 5.NF.B.4.

Directions: Show ALL work by hand in your math notebook (graph paper is best). Label every answer with units. Figures are not drawn to scale.

Section A · Area Is Multiplication (Day 5)

1. A rectangle is 23 ft long and 14 ft wide. Find its area.

Answer: _____

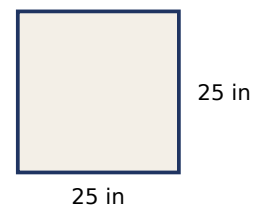


2. Two Nerf darts landed on 36 and 18. Use them as the length and width (in cm) of a rectangle. What is the area?

Answer: _____

3. A square tile has sides of 25 in. What is its area?

Answer: _____

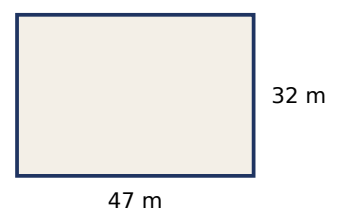


4. A classroom rug is 12 ft by 9 ft. How many 1-foot-by-1-foot squares would it take to cover the rug exactly?

Answer: _____

5. A community garden is 47 m long and 32 m wide. Find the area.

Answer: _____



6. Find the mistake. Jordan found the area of a 15 by 13 rectangle like this: $15 \times 10 = 150$, then $150 + 3 = 153$. What went wrong? What is the correct area?

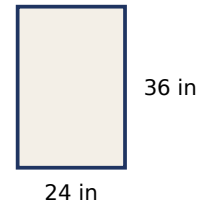
Answer: _____

7. Which rectangle has the greater area: 18 by 16 or 17 by 17? By how much?

Answer: _____

8. A poster is 24 in by 36 in. Find its area, and explain what the unit "square inches" means.

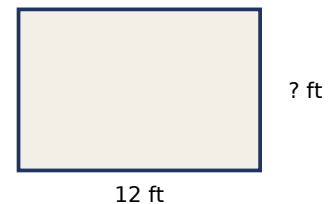
Answer: _____



Section B · Area Unlocks the Missing Side (Day 6)

9. A rectangle has an area of 96 sq ft. Its length is 12 ft. Find the width.

Answer: _____



10. A rectangle has an area of 391 sq m and a width of 17 m. Find the length.

Answer: _____

11. A square has an area of 144 sq in. How long is each side?

Answer: _____

12. Square vs. rectangle repair. Maya says, "Every rectangle is a square." Is she right? Explain and give an example.

Answer: _____

13. A rectangle has an area of 252 sq in and one side of 14 in. Find the other side, then find the perimeter.

Answer: _____

14. A square has an area of 81 sq cm. What is its perimeter?

Answer: _____

15. A rectangular gym floor has an area of 480 sq ft and is 24 ft long. How wide is it?

Answer: _____

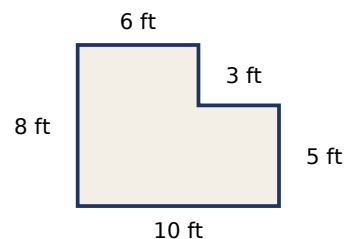
16. Two rectangles each have an area of 60 sq units. One has a side of 5; the other has a side of 4. Find each missing side. Which rectangle has the larger perimeter?

Answer: _____

Section C · Area of Compound Rectangles (Day 7)

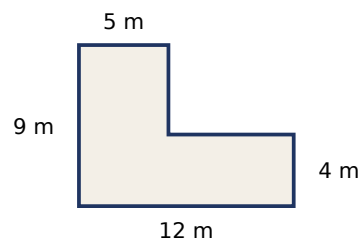
17. Find the area of the L-shaped floor by splitting it into two rectangles. (All angles are right angles.)

Answer: _____



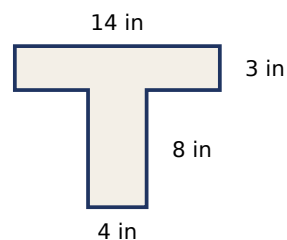
18. Find the area of this L-shape TWO ways: (a) split and add, (b) enclose and subtract. Show both give the same answer.

Answer: _____



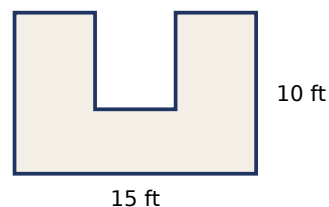
19. Find the area of the T-shape. The top bar is 14 in wide and 3 in tall. The stem is 4 in wide and 8 in tall.

Answer: _____



20. A 15 ft by 10 ft rectangle has a 5 ft wide, 6 ft deep notch cut out of the top middle. Find the area of the shape that remains.

Answer: _____

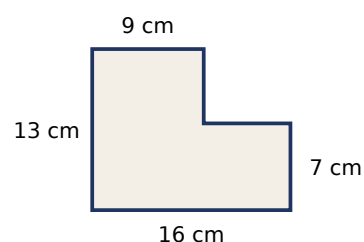


21. A bedroom is 20 ft by 15 ft. A 4 ft by 6 ft closet in the corner will NOT get carpet. Carpet costs \$3 per square foot. How much will the carpet cost?

Answer: _____

22. The two unlabeled side lengths are missing. Find them, then find the total area.

Answer: _____

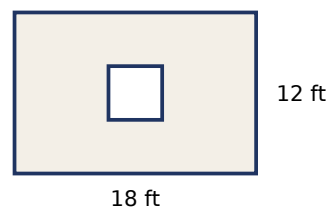


23. Find the mistake. For the floor in Question 17 (10 by 8 with a 4 by 3 notch), Deshawn wrote $10 \times 5 + 6 \times 8 = 50 + 48 = 98$ sq ft. What went wrong?

Answer: _____

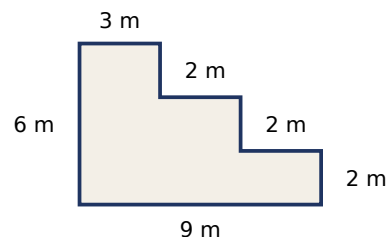
24. A rectangular patio is 18 ft by 12 ft. A 4 ft by 4 ft square flower bed in the middle will not be paved. How many square feet will be paved?

Answer: _____



25. A staircase shape is made of three stacked rectangles, each 2 m tall. Their widths are 9 m, 6 m, and 3 m. Find the total area.

Answer: _____



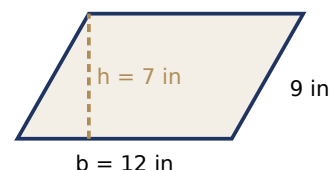
26. A T-shaped stage has a top section 20 ft by 6 ft and a stem 8 ft by 10 ft. It will be covered with square tiles that are 2 ft by 2 ft. How many tiles are needed?

Answer: _____

Section D · Area of a Parallelogram (Days 8-9)

27. Find the area of the parallelogram. (The dashed line is the height.)

Answer: _____



28. A parallelogram has a base of 15 cm and a height of 8 cm. Find its area.

Answer: _____

29. A parallelogram has an area of 126 sq in and a base of 14 in. Find its height.

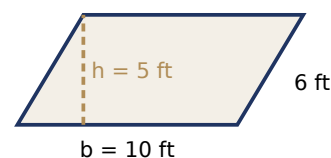
Answer: _____

30. A parallelogram has an area of 90 sq m and a height of 6 m. Find its base.

Answer: _____

31. Find the mistake. A parallelogram has base 10 ft, slanted side 6 ft, and height 5 ft. A student said the area is $10 \times 6 = 60$ sq ft. Explain the error and find the correct area.

Answer: _____



32. Explain why the area of a parallelogram is base $\times$ height. Use the idea of cutting and sliding.

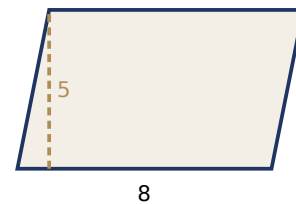
Answer: _____

33. A parallelogram-shaped garden has a base of 20 ft and a height of 13 ft. A 3 ft by 3 ft square pond sits inside it. How much area is left for plants?

Answer: _____

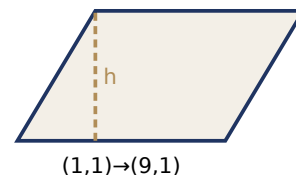
34. Two parallelograms both have a base of 8 cm and a height of 5 cm, but one leans much farther than the other. Compare their areas.

Answer: _____



35. On a coordinate grid, a parallelogram has vertices (1, 1), (9, 1), (12, 6), and (4, 6). Find its area.

Answer: _____



36. A rectangle is 9 in by 6 in. A parallelogram has a base of 9 in, a slanted side of 6 in, and a height of 4 in. Which has more area? How much more?

Answer: _____

Section E • Skill Breakout: Multiplying Decimals (Day 10)

37. Multiply: 0.6×0.4

Answer: _____

38. Multiply: 3.2×1.5

Answer: _____

39. Multiply: 2.05×4

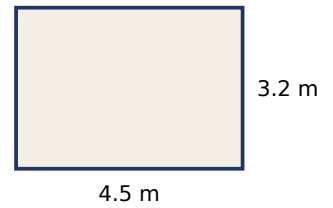
Answer: _____

40. Multiply: 12.5×0.08

Answer: _____

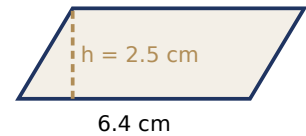
41. A rectangular rug is 4.5 m by 3.2 m. Find its area.

Answer: _____



42. A parallelogram has a base of 6.4 cm and a height of 2.5 cm. Find its area.

Answer: _____



43. Find the mistake. Priya says $0.3 \times 0.2 = 0.6$. Explain her error and give the correct product.

Answer: _____

Section F • Skill Breakout: Multiplying Fractions (Day 10)

44. Multiply: $\frac{2}{3} \times \frac{3}{4}$. Simplify.

Answer: _____

45. Find $\frac{3}{5}$ of 20.

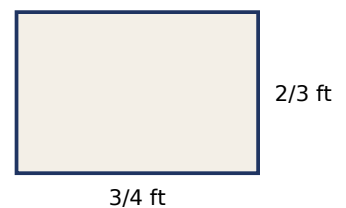
Answer: _____

46. Multiply: $1\frac{1}{2} \times 2\frac{2}{3}$

Answer: _____

47. A rectangle is $\frac{3}{4}$ ft long and $\frac{2}{3}$ ft wide. Find its area.

Answer: _____



48. Same function, different notation. Compute 0.5×36 and $\frac{1}{2} \times 36$. Then compute 0.25×48 and $\frac{1}{4} \times 48$. What do you notice? Explain.

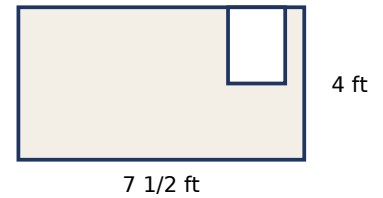
Answer: _____

49. A parallelogram has a base of $5 \frac{1}{2}$ in and a height of 4 in. Find its area.

Answer: _____

50. Challenge. A rectangular yard is $7 \frac{1}{2}$ ft by 4 ft. A shed that is 1.5 ft by 2 ft sits in the yard. How many square feet of yard are NOT covered by the shed?

Answer: _____



Answer Key & In-Depth Solutions

Day 11 Review · Days 5–10 · Teacher / family reference — try every problem before checking!

Section A · Area Is Multiplication (Day 5)

1. A rectangle is 23 ft long and 14 ft wide. Find its area.

- Step 1: Area of a rectangle = length $\times$ width, so we need 23×14 .
- Step 2: Break both factors apart (area model): $23 = 20 + 3$ and $14 = 10 + 4$.
- Step 3: Multiply every part: $20 \times 10 = 200$, $20 \times 4 = 80$, $3 \times 10 = 30$, $3 \times 4 = 12$.
- Step 4: Add the partial products: $200 + 80 + 30 + 12 = 322$.
- Step 5: Units: feet $\times$ feet = square feet. Area = **322 sq ft**.

✓ **Answer: 322 sq ft**

2. Two Nerf darts landed on 36 and 18. Use them as the length and width (in cm) of a rectangle. What is the area?

- Step 1: Area = 36×18 .
- Step 2: Area model: $36 = 30 + 6$ and $18 = 10 + 8$.
- Step 3: $30 \times 10 = 300$, $30 \times 8 = 240$, $6 \times 10 = 60$, $6 \times 8 = 48$.
- Step 4: $300 + 240 + 60 + 48 = 648$.
- Step 5: Check with the standard algorithm: $36 \times 8 = 288$ and $36 \times 10 = 360$; $288 + 360 = 648$. ✓
- Step 6: Area = **648 sq cm**.

✓ **Answer: 648 sq cm**

3. A square tile has sides of 25 in. What is its area?

- Step 1: A square is a rectangle with all sides equal, so Area = side $\times$ side = 25×25 .
- Step 2: $25 \times 25 = (20 + 5)(20 + 5) = 400 + 100 + 100 + 25 = 625$.
- Step 3: Area = **625 sq in**.

✓ **Answer: 625 sq in**

4. A classroom rug is 12 ft by 9 ft. How many 1-foot-by-1-foot squares would it take to cover the rug exactly?

- Step 1: Each 1 ft $\times$ 1 ft square covers exactly 1 square foot, so the number of squares equals the area.
- Step 2: Think of rows: there are 9 rows with 12 squares in each row.
- Step 3: $12 \times 9 = 108$.
- Step 4: It takes **108 squares** (108 sq ft).

✓ **Answer: 108 squares**

5. A community garden is 47 m long and 32 m wide. Find the area.

- Step 1: Area = 47×32 .
- Step 2: Area model: $47 = 40 + 7$ and $32 = 30 + 2$.
- Step 3: $40 \times 30 = 1,200$; $40 \times 2 = 80$; $7 \times 30 = 210$; $7 \times 2 = 14$.
- Step 4: $1,200 + 80 + 210 + 14 = 1,504$.
- Step 5: Estimate check: $50 \times 30 = 1,500$, so 1,504 is reasonable.
- Step 6: Area = **1,504 sq m**.

✓ **Answer: 1,504 sq m**

6. Find the mistake. Jordan found the area of a 15 by 13 rectangle like this: $15 \times 10 = 150$, then $150 + 3 = 153$. What went wrong? What is the correct area?

- Step 1: $13 = 10 + 3$, so Jordan must multiply 15 by BOTH parts.
- Step 2: Jordan did $15 \times 10 = 150$ correctly, but then only added 3 instead of multiplying 15×3 .
- Step 3: The missing partial product is $15 \times 3 = 45$.
- Step 4: Correct area: $150 + 45 = 195$.
- Step 5: Area = **195 square units**. (Lesson: every part of one factor gets multiplied by every part of the other.)

✓ **Answer: 195 square units**

7. Which rectangle has the greater area: 18 by 16 or 17 by 17? By how much?

- Step 1: 18×16 : $(10 + 8)(10 + 6) = 100 + 60 + 80 + 48 = 288$.
- Step 2: 17×17 : $(10 + 7)(10 + 7) = 100 + 70 + 70 + 49 = 289$.
- Step 3: $289 - 288 = 1$.
- Step 4: The **17 by 17 square** is greater by **1 square unit**. Both have the same perimeter (68), but the square shape holds a little more area.

✓ **Answer: 17 × 17 is greater, by 1 square unit**

8. A poster is 24 in by 36 in. Find its area, and explain what the unit "square inches" means.

- Step 1: Area = 24×36 .
- Step 2: $24 \times 36 = (20 + 4)(30 + 6) = 600 + 120 + 120 + 24 = 864$.
- Step 3: "Square inches" means we are counting how many 1-inch-by-1-inch squares cover the poster.
- Step 4: Area = **864 sq in**: 864 tiny 1-inch squares would cover it exactly.

✓ **Answer: 864 sq in**

Section B · Area Unlocks the Missing Side (Day 6)

9. A rectangle has an area of 96 sq ft. Its length is 12 ft. Find the width.

- Step 1: Area = length $\times$ width, so $96 = 12 \times \text{width}$.
- Step 2: Run area in reverse: width = area $\div$ length = $96 \div 12$.
- Step 3: $12 \times 8 = 96$, so $96 \div 12 = 8$.
- Step 4: Width = **8 ft**. Check: $12 \times 8 = 96$. ✓

✓ **Answer: 8 ft**

10. A rectangle has an area of 391 sq m and a width of 17 m. Find the length.

- Step 1: Length = $391 \div 17$.
- Step 2: Estimate: $17 \times 20 = 340$, which leaves $391 - 340 = 51$.
- Step 3: $17 \times 3 = 51$, so we need $20 + 3 = 23$ groups of 17.
- Step 4: Length = **23 m**. Check: $17 \times 23 = 340 + 51 = 391$. ✓

✓ **Answer: 23 m**

11. A square has an area of 144 sq in. How long is each side?

- Step 1: In a square, both sides are the same number, so we need a number times itself that equals 144.
- Step 2: Try: $10 \times 10 = 100$ (too small), $12 \times 12 = 144$. ✓
- Step 3: Each side = **12 in**.

✓ **Answer: 12 in**

12. Square vs. rectangle repair. Maya says, "Every rectangle is a square." Is she right? Explain and give an example.

- Step 1: A rectangle is a four-sided shape with four right angles.
- Step 2: A square is a rectangle that ALSO has all four sides equal.
- Step 3: So a square is a special kind of rectangle, but a rectangle does not have to have equal sides.
- Step 4: Counterexample: a 4 by 6 rectangle has four right angles but its sides are not all equal, so it is **not** a square.
- Step 5: Maya has it backwards: **every square is a rectangle, but not every rectangle is a square.**

✓ **Answer: No — every square is a rectangle, but not every rectangle is a square.**

13. A rectangle has an area of 252 sq in and one side of 14 in. Find the other side, then find the perimeter.

- Step 1: Other side = $252 \div 14$.
- Step 2: $14 \times 10 = 140 \rightarrow 252 - 140 = 112$. $14 \times 8 = 112$. So $10 + 8 = 18$.
- Step 3: The other side is 18 in. Check: $14 \times 18 = 252$. ✓
- Step 4: Perimeter = $2 \times (14 + 18) = 2 \times 32 = 64$.
- Step 5: Other side = **18 in**; perimeter = **64 in**. (Area is in square inches, perimeter is in plain inches.)

✓ **Answer: 18 in; perimeter 64 in**

14. A square has an area of 81 sq cm. What is its perimeter?

- Step 1: Find the side first: which number times itself is 81? $9 \times 9 = 81$, so the side is 9 cm.
- Step 2: A square has 4 equal sides: perimeter = $4 \times 9 = 36$.
- Step 3: Perimeter = **36 cm**.

✓ **Answer: 36 cm**

15. A rectangular gym floor has an area of 480 sq ft and is 24 ft long. How wide is it?

- Step 1: Width = $480 \div 24$.
- Step 2: Think: $24 \times 2 = 48$, so $24 \times 20 = 480$.
- Step 3: Width = **20 ft**. Check: $24 \times 20 = 480$. ✓

✓ **Answer: 20 ft**

16. Two rectangles each have an area of 60 sq units. One has a side of 5; the other has a side of 4. Find each missing side. Which rectangle has the larger perimeter?

- Step 1: Rectangle 1: $60 \div 5 = 12$, so it is 5 by 12.
- Step 2: Rectangle 2: $60 \div 4 = 15$, so it is 4 by 15.
- Step 3: Perimeter 1: $2 \times (5 + 12) = 34$. Perimeter 2: $2 \times (4 + 15) = 38$.
- Step 4: Missing sides: **12 and 15**. The **4 by 15** rectangle has the larger perimeter ($38 > 34$), even though the areas are equal.

✓ **Answer: 12 and 15; the 4 × 15 rectangle (38 vs. 34)**

Section C • Area of Compound Rectangles (Day 7)

17. Find the area of the L-shaped floor by splitting it into two rectangles. (All angles are right angles.)

- Step 1: Find any hidden sides: the notch is $10 - 6 = 4$ ft wide and $8 - 5 = 3$ ft tall.
- Step 2: Split with a vertical cut at 6 ft. Left rectangle: $6 \times 8 = 48$.
- Step 3: Right rectangle: 4 ft wide ($10 - 6$) and 5 ft tall $\rightarrow 4 \times 5 = 20$.
- Step 4: Add: $48 + 20 = 68$.
- Step 5: Area = **68 sq ft**.

✓ **Answer: 68 sq ft**

18. Find the area of this L-shape TWO ways: (a) split and add, (b) enclose and subtract. Show both give the same answer.

- Step 1: **(a) Split and add:** cut horizontally at height 4. Bottom rectangle: $12 \times 4 = 48$. Top rectangle: 5 wide $\times (9 - 4) = 5 \times 5 = 25$. Total $48 + 25 = 73$.
- Step 2: **(b) Enclose and subtract:** the enclosing rectangle is $12 \times 9 = 108$.
- Step 3: The missing (empty) piece is $(12 - 5)$ wide by $(9 - 4)$ tall $= 7 \times 5 = 35$.
- Step 4: $108 - 35 = 73$.
- Step 5: Both methods give **73 sq m**. ✓

✓ **Answer: 73 sq m**

19. Find the area of the T-shape. The top bar is 14 in wide and 3 in tall. The stem is 4 in wide and 8 in tall.

- Step 1: Split into the top bar and the stem.
- Step 2: Top bar: $14 \times 3 = 42$.
- Step 3: Stem: $4 \times 8 = 32$.
- Step 4: Add: $42 + 32 = 74$.
- Step 5: Check by subtracting: enclosing rectangle $14 \times 11 = 154$. Each empty corner is $5 \times 8 = 40$ (because $(14 - 4) \div 2 = 5$). $154 - 40 - 40 = 74$. ✓
- Step 6: Area = **74 sq in**.

✓ **Answer: 74 sq in**

20. A 15 ft by 10 ft rectangle has a 5 ft wide, 6 ft deep notch cut out of the top middle. Find the area of the shape that remains.

- Step 1: Enclose and subtract is fastest here.
- Step 2: Whole rectangle: $15 \times 10 = 150$.
- Step 3: Notch (the missing piece): $5 \times 6 = 30$.
- Step 4: $150 - 30 = 120$.
- Step 5: Area = **120 sq ft**.

✓ **Answer: 120 sq ft**

21. A bedroom is 20 ft by 15 ft. A 4 ft by 6 ft closet in the corner will NOT get carpet. Carpet costs \$3 per square foot. How much will the carpet cost?

- Step 1: Room area: $20 \times 15 = 300$ sq ft.
- Step 2: Closet area: $4 \times 6 = 24$ sq ft.
- Step 3: Carpeted area: $300 - 24 = 276$ sq ft.
- Step 4: Cost: $276 \times \$3 = \828 ($200 \times 3 = 600$, $70 \times 3 = 210$, $6 \times 3 = 18$; $600 + 210 + 18 = 828$).
- Step 5: The carpet costs **\$828**.

✓ **Answer: \$828**

22. The two unlabeled side lengths are missing. Find them, then find the total area.

- Step 1: Missing horizontal side: the bottom is 16 and the top is 9, so $16 - 9 = 7$ cm.
- Step 2: Missing vertical side: the left is 13 and the right is 7, so $13 - 7 = 6$ cm.
- Step 3: Split horizontally at height 7: bottom $16 \times 7 = 112$; top $9 \times 6 = 54$.
- Step 4: $112 + 54 = 166$.
- Step 5: Check: $16 \times 13 = 208$; empty corner $7 \times 6 = 42$; $208 - 42 = 166$. ✓
- Step 6: Missing sides: **7 cm and 6 cm**; area = **166 sq cm**.

✓ **Answer: 7 cm and 6 cm; area 166 sq cm**

23. Find the mistake. For the floor in Question 17 (10 by 8 with a 4 by 3 notch), Deshawn wrote $10 \times 5 + 6 \times 8 = 50 + 48 = 98$ sq ft. What went wrong?

- Step 1: The 10×5 rectangle is the bottom strip. The 6×8 rectangle is the whole left side.
- Step 2: These two rectangles OVERLAP in the bottom-left corner: a $6 \times 5 = 30$ sq ft region got counted twice.
- Step 3: Fix: $98 - 30 = 68$, OR use non-overlapping pieces: $6 \times 8 + 4 \times 5 = 48 + 20 = 68$.
- Step 4: Rule: when you split a shape, the pieces must not overlap. Correct area = **68 sq ft**.

✓ **Answer: He counted the 6×5 overlap twice; correct area is 68 sq ft**

24. A rectangular patio is 18 ft by 12 ft. A 4 ft by 4 ft square flower bed in the middle will not be paved. How many square feet will be paved?

- Step 1: Patio area: $18 \times 12 = 216$ ($18 \times 10 = 180$, $18 \times 2 = 36$).
- Step 2: Flower bed: $4 \times 4 = 16$.
- Step 3: $216 - 16 = 200$.
- Step 4: Paved area = **200 sq ft**.

✓ **Answer: 200 sq ft**

25. A staircase shape is made of three stacked rectangles, each 2 m tall. Their widths are 9 m, 6 m, and 3 m. Find the total area.

- Step 1: Bottom step: $9 \times 2 = 18$.
- Step 2: Middle step: $6 \times 2 = 12$.
- Step 3: Top step: $3 \times 2 = 6$.
- Step 4: Add: $18 + 12 + 6 = 36$.
- Step 5: Check: enclosing rectangle $9 \times 6 = 54$; empty pieces $3 \times 2 = 6$ and $6 \times 2 = 12$; $54 - 18 = 36$. ✓
- Step 6: Area = **36 sq m**.

✓ **Answer: 36 sq m**

26. A T-shaped stage has a top section 20 ft by 6 ft and a stem 8 ft by 10 ft. It will be covered with square tiles that are 2 ft by 2 ft. How many tiles are needed?

- Step 1: Top section: $20 \times 6 = 120$ sq ft. Stem: $8 \times 10 = 80$ sq ft.
- Step 2: Total stage area: $120 + 80 = 200$ sq ft.
- Step 3: Each tile covers $2 \times 2 = 4$ sq ft (NOT 2 sq ft!).
- Step 4: Number of tiles: $200 \div 4 = 50$.
- Step 5: They need **50 tiles**.

✓ **Answer: 50 tiles**

Section D · Area of a Parallelogram (Days 8-9)

27. Find the area of the parallelogram. (The dashed line is the height.)

- Step 1: Area of a parallelogram = base $\times$ height (the height is perpendicular to the base).
- Step 2: Base = 12 in; height = 7 in. The slanted side (9 in) is NOT the height — ignore it.
- Step 3: $12 \times 7 = 84$.
- Step 4: Area = **84 sq in**.

✓ **Answer: 84 sq in**

28. A parallelogram has a base of 15 cm and a height of 8 cm. Find its area.

- Step 1: $A = b \times h = 15 \times 8$.
- Step 2: $15 \times 8 = (10 \times 8) + (5 \times 8) = 80 + 40 = 120$.
- Step 3: Area = **120 sq cm**.

✓ **Answer: 120 sq cm**

29. A parallelogram has an area of 126 sq in and a base of 14 in. Find its height.

- Step 1: $A = b \times h$, so $126 = 14 \times h$.
- Step 2: $h = 126 \div 14$.
- Step 3: $14 \times 9 = 126$, so $h = 9$.
- Step 4: Height = **9 in**. Check: $14 \times 9 = 126$. ✓

✓ **Answer: 9 in**

30. A parallelogram has an area of 90 sq m and a height of 6 m. Find its base.

- Step 1: $90 = b \times 6$, so $b = 90 \div 6$.
- Step 2: $6 \times 15 = 90$.
- Step 3: Base = **15 m**.

✓ **Answer: 15 m**

31. Find the mistake. A parallelogram has base 10 ft, slanted side 6 ft, and height 5 ft. A student said the area is $10 \times 6 = 60$ sq ft. Explain the error and find the correct area.

- Step 1: The student used base $\times$ SLANT. The formula needs base $\times$ HEIGHT.
- Step 2: The slanted side leans over, so it is longer than the straight-up height ($6 > 5$). Using it makes the area too big.
- Step 3: Picture it: cut the triangle off one end and slide it to the other — the new rectangle is 10 long and 5 tall, not 6 tall.
- Step 4: Correct area: $10 \times 5 = 50$.
- Step 5: Area = **50 sq ft**.

✓ **Answer: 50 sq ft**

32. Explain why the area of a parallelogram is base $\times$ height. Use the idea of cutting and sliding.

- Step 1: Draw the height from one top corner straight down to the base. This cuts off a right triangle.
- Step 2: Slide that triangle to the opposite end. It fits exactly into the gap.
- Step 3: The new shape is a rectangle. No area was added or removed — we only moved a piece.
- Step 4: The rectangle's length is the parallelogram's base, and its width is the parallelogram's height.
- Step 5: Rectangle area = length $\times$ width, so **parallelogram area = base $\times$ height**.

✓ **Answer: It rearranges into a rectangle with the same base and height.**

33. A parallelogram-shaped garden has a base of 20 ft and a height of 13 ft. A 3 ft by 3 ft square pond sits inside it. How much area is left for plants?

- Step 1: Garden: $20 \times 13 = 260$ sq ft.
- Step 2: Pond: $3 \times 3 = 9$ sq ft.
- Step 3: $260 - 9 = 251$.
- Step 4: Planting area = **251 sq ft**.

✓ **Answer: 251 sq ft**

34. Two parallelograms both have a base of 8 cm and a height of 5 cm, but one leans much farther than the other. Compare their areas.

- Step 1: Area depends only on base and height: $8 \times 5 = 40$ for each.
- Step 2: How far it leans changes the slanted side length, but not the height.
- Step 3: Both areas = **40 sq cm**. They are equal.

✓ **Answer: Both are 40 sq cm — equal.**

35. On a coordinate grid, a parallelogram has vertices (1, 1), (9, 1), (12, 6), and (4, 6). Find its area.

- Step 1: The bottom side runs from (1, 1) to (9, 1): base = $9 - 1 = 8$ units.
- Step 2: The top side is at $y = 6$ and the base is at $y = 1$, so height = $6 - 1 = 5$ units (count straight up, not along the slant).
- Step 3: $A = 8 \times 5 = 40$.
- Step 4: Area = **40 square units**.

✓ **Answer: 40 square units**

36. A rectangle is 9 in by 6 in. A parallelogram has a base of 9 in, a slanted side of 6 in, and a height of 4 in. Which has more area? How much more?

- Step 1: Rectangle: $9 \times 6 = 54$ sq in.
- Step 2: Parallelogram: base $\times$ height = $9 \times 4 = 36$ sq in (not 9×6 !).
- Step 3: $54 - 36 = 18$.
- Step 4: The **rectangle** has **18 sq in** more, even though the side lengths match.

✓ **Answer: Rectangle, by 18 sq in**

Section E • Skill Breakout: Multiplying Decimals (Day 10)

37. Multiply: 0.6×0.4

- Step 1: Ignore the decimals and multiply the whole numbers: $6 \times 4 = 24$.
- Step 2: Count decimal places in the factors: 0.6 has 1, 0.4 has 1 $\rightarrow$ 2 total.
- Step 3: Place the decimal so the answer has 2 decimal places: 0.24.
- Step 4: Meaning: 6 tenths $\times$ 4 tenths = 24 hundredths. Answer: **0.24**.

✓ **Answer: 0.24**

38. Multiply: 3.2×1.5

- Step 1: Multiply as whole numbers: $32 \times 15 = 32 \times 10 + 32 \times 5 = 320 + 160 = 480$.
- Step 2: Decimal places: $1 + 1 = 2$, so $480 \rightarrow 4.80$.
- Step 3: $4.80 = 4.8$. Estimate check: $3 \times 1.5 = 4.5$, close to 4.8. ✓
- Step 4: Answer: **4.8**.

✓ **Answer: 4.8**

39. Multiply: 2.05×4

- Step 1: $205 \times 4 = 820$.
- Step 2: Decimal places: 2 (from 2.05) + 0 = 2, so $820 \rightarrow 8.20$.
- Step 3: Estimate: $2 \times 4 = 8$. ✓
- Step 4: Answer: **8.2**.

✓ **Answer: 8.2**

40. Multiply: 12.5×0.08

- Step 1: $125 \times 8 = 1,000$.
- Step 2: Decimal places: 1 (12.5) + 2 (0.08) = 3, so $1000 \rightarrow 1.000$.
- Step 3: $1.000 = 1$. Estimate: 12×0.08 is about $12 \times 0.1 = 1.2$, so about 1. ✓
- Step 4: Answer: **1**.

✓ **Answer: 1**

41. A rectangular rug is 4.5 m by 3.2 m. Find its area.

- Step 1: Area = 4.5×3.2 .
- Step 2: $45 \times 32 = 45 \times 30 + 45 \times 2 = 1,350 + 90 = 1,440$.
- Step 3: Decimal places: $1 + 1 = 2 \rightarrow 14.40$.
- Step 4: Estimate: $4.5 \times 3 = 13.5$, close. ✓
- Step 5: Area = **14.4 sq m**.

✓ **Answer: 14.4 sq m**

42. A parallelogram has a base of 6.4 cm and a height of 2.5 cm. Find its area.

- Step 1: $A = b \times h = 6.4 \times 2.5$.
- Step 2: $64 \times 25 = 1,600$ (because $4 \times 25 = 100$, so $64 \times 25 = 16 \times 100$).
- Step 3: Decimal places: $1 + 1 = 2 \rightarrow 16.00$.
- Step 4: Area = **16 sq cm**.

✓ **Answer: 16 sq cm**

43. Find the mistake. Priya says $0.3 \times 0.2 = 0.6$. Explain her error and give the correct product.

- Step 1: $3 \times 2 = 6$ is correct, but Priya did not count decimal places.
- Step 2: 0.3 has 1 decimal place and 0.2 has 1, so the product needs 2: 0.06.
- Step 3: Reasoning check: 0.3×0.2 means "3 tenths OF 2 tenths." Taking part of a number smaller than 1 gives something even smaller — 0.6 is bigger than both factors, so it cannot be right.
- Step 4: Correct product: **0.06**.

✓ **Answer: 0.06**

Section F • Skill Breakout: Multiplying Fractions (Day 10)

44. Multiply: $\frac{2}{3} \times \frac{3}{4}$. Simplify.

- Step 1: Multiply numerators: $2 \times 3 = 6$. Multiply denominators: $3 \times 4 = 12$.
- Step 2: $\frac{6}{12}$ simplifies: divide top and bottom by 6 $\rightarrow \frac{1}{2}$.
- Step 3: Shortcut: cross-simplify first — the 3s cancel, and $\frac{2}{4} = \frac{1}{2}$.
- Step 4: Answer: **$\frac{1}{2}$** .

✓ **Answer: $\frac{1}{2}$**

45. Find $\frac{3}{5}$ of 20.

- Step 1: "Of" means multiply: $\frac{3}{5} \times 20$.
- Step 2: Find $\frac{1}{5}$ of 20 first: $20 \div 5 = 4$.
- Step 3: Then $\frac{3}{5}$ is three of those: $3 \times 4 = 12$.
- Step 4: Answer: **12**.

✓ **Answer: 12**

46. Multiply: $1\frac{1}{2} \times 2\frac{2}{3}$

- Step 1: Change mixed numbers to improper fractions: $1\frac{1}{2} = \frac{3}{2}$ and $2\frac{2}{3} = \frac{8}{3}$.
- Step 2: $\frac{3}{2} \times \frac{8}{3} = \frac{24}{6}$.
- Step 3: $\frac{24}{6} = 4$.
- Step 4: Estimate: about $1.5 \times 2.7 \approx 4$. ✓ Answer: **4**.

✓ **Answer: 4**

47. A rectangle is $\frac{3}{4}$ ft long and $\frac{2}{3}$ ft wide. Find its area.

- Step 1: Area = $\frac{3}{4} \times \frac{2}{3} = \frac{6}{12}$.
- Step 2: $\frac{6}{12} = \frac{1}{2}$.
- Step 3: This makes sense: the rectangle is smaller than 1 ft by 1 ft, so its area should be less than 1 sq ft.
- Step 4: Area = **$\frac{1}{2}$ sq ft**.

✓ **Answer: $\frac{1}{2}$ sq ft**

48. Same function, different notation. Compute 0.5×36 and $\frac{1}{2} \times 36$. Then compute 0.25×48 and $\frac{1}{4} \times 48$. What do you notice? Explain.

- Step 1: 0.5×36 : $5 \times 36 = 180$, one decimal place $\rightarrow 18.0 = 18$.
- Step 2: $\frac{1}{2} \times 36 = 36 \div 2 = 18$.
- Step 3: 0.25×48 : $25 \times 48 = 1,200$, two decimal places $\rightarrow 12.00 = 12$.
- Step 4: $\frac{1}{4} \times 48 = 48 \div 4 = 12$.
- Step 5: Because $0.5 = \frac{1}{2}$ and $0.25 = \frac{1}{4}$, each pair is the SAME multiplication written two ways.
- Step 6: Results: **$18 = 18$ and $12 = 12$** . Pick whichever notation makes the math easier.

✓ **Answer: 18 and 18; 12 and 12 — the decimal and fraction name the same number.**

49. A parallelogram has a base of $5\frac{1}{2}$ in and a height of 4 in. Find its area.

- Step 1: $A = b \times h = 5\frac{1}{2} \times 4$.
- Step 2: Distribute: $5 \times 4 = 20$ and $\frac{1}{2} \times 4 = 2$.
- Step 3: $20 + 2 = 22$. (Or: $11\frac{1}{2} \times 4 = 44\frac{1}{2} = 22$.)
- Step 4: Area = **22 sq in**.

✓ **Answer: 22 sq in**

50. Challenge. A rectangular yard is $7\frac{1}{2}$ ft by 4 ft. A shed that is 1.5 ft by 2 ft sits in the yard. How many square feet of yard are NOT covered by the shed?

- Step 1: Yard: $7\frac{1}{2} \times 4 = 28 + 2 = 30$ sq ft ($7 \times 4 = 28$ and $\frac{1}{2} \times 4 = 2$).
- Step 2: Shed: $1.5 \times 2 = 3$ sq ft ($15 \times 2 = 30$, one decimal place $\rightarrow 3.0$).
- Step 3: Uncovered area (enclose and subtract): $30 - 3 = 27$.
- Step 4: Notice $7\frac{1}{2}$ and 1.5 use different notation, but the multiplication works the same way.
- Step 5: Uncovered area = **27 sq ft**.

✓ **Answer: 27 sq ft**